# Boundary-Preserved Deterministic QEM

# Out-of-Core Simplification for Large Triangle Meshes

**Publisher:** Make Your Own World
**Version:** 1.0

## Abstract

Large triangle meshes can exceed practical working-memory limits even when their binary STL files fit on disk. This report presents a bounded-memory pipeline that partitions a binary STL into deterministic adaptive octree tiles, simplifies tiles concurrently, freezes partition boundaries, checkpoints every accepted result, and streams a final STL without constructing a global vertex or face array.

On a 4,126,315-triangle reference mesh, eight CPU workers completed the tiled reduction in 4.67 seconds compared with 27.34 seconds for one worker, a 5.85x speedup. Sequential, parallel, and cached runs produced byte-identical 249,679-triangle outputs. The requested count was 249,184. Boundary preservation retained 495 triangles. Connected-surface tests measured zero assembly loss and no new open partition seams.

## Contributions

The contribution is the integrated production contract: singular triangle ownership, stable allocation, immutable float32 partition coordinates, memory-bounded scheduling, atomic content-validated checkpoints, cancellation-safe streamed assembly, and byte-level determinism. Octrees, Morton order, QEM, locked boundaries, and external-memory processing are established foundations and are attributed accordingly.

# 1. The coupled problem

A compact triangle stream expands during reduction into welded vertices, indexed faces, adjacency, quadrics, candidate contractions, and temporary native state. Peak memory is therefore the sum of simultaneous structures, not the input file size:

$$M_{\text{peak}} \approx M_{\text{source}} + M_{\text{index}} + M_{\text{adjacency}} + M_{\text{quadrics}} + M_{\text{candidates}} + M_{\text{temporary}}$$

$\mathbf{M_{peak}}$ is peak resident geometry memory; $\mathbf{M_{source}}$ is the streamed source state; $\mathbf{M_{index}}$ is the spatial index; $\mathbf{M_{adjacency}}$ is connectivity state; $\mathbf{M_{quadrics}}$ stores error matrices; $\mathbf{M_{candidates}}$ stores proposed collapses; and $\mathbf{M_{temporary}}$ is native working memory. All terms use bytes.

M_peak is peak resident geometry memory. Every term is live during at least part of an in-memory reduction. Streaming the reader alone does not solve the problem when topology and contraction state still scale with the complete mesh.

Partitioning introduces six linked challenges: one owner for every source triangle, continuous seams, a deterministic global target, bounded concurrency, validated recovery, and stable output order. Fixing one while ignoring the others produces a reducer that is fast in a narrow test but unsafe for long production runs.

**Boundary-Preserved Deterministic QEM pipeline**



## 2. Local QEM mathematics

For plane p = (a,b,c,d)^T and homogeneous vertex v_bar = (x,y,z,1)^T, the plane quadric and accumulated vertex quadric are:

$$Q_p = pp^T, \qquad Q_v = \sum_{p \in P(v)} Q_p$$

**p** is a normalized incident face plane in homogeneous form; $\mathbf{Q_p}$ is that plane's 4-by-4 error matrix; **P(v)** is the set of planes incident on vertex **v**; and $\mathbf{Q_v}$ is the accumulated vertex quadric.

The contraction cost for vertices v1 and v2 at candidate v'_bar is:

$$E(\bar{\mathbf{v}}') = \bar{\mathbf{v}}'^T (Q_{v_1} + Q_{v_2}) \bar{\mathbf{v}}'$$

$\mathbf{v}'$ is the proposed homogeneous replacement vertex; $\mathbf{Q_{v1}}$ and $\mathbf{Q_{v2}}$ are the endpoint quadrics; and **E** is the estimated squared plane-distance error used to rank the contraction.

p is an incident face plane. P(v) is the set of planes incident on v. Q_v accumulates squared distance penalties. E is the estimated geometric error introduced by the candidate contraction. The native local reducer performs these contractions. The method described here supplies safe ownership, constraints, scheduling, recovery, and assembly around it.

## 3. Spatial ownership

For triangle f with vertices x_f1, x_f2, and x_f3, its persistent owner is selected from its centroid:

$$\mathbf{c}_f = \frac{\mathbf{x}_{f,1} + \mathbf{x}_{f,2} + \mathbf{x}_{f,3}}{3}$$

**f** is one source triangle; $\mathbf{x_{f,1}}$, $\mathbf{x_{f,2}}$, and $\mathbf{x_{f,3}}$ are its vertex positions; and $\mathbf{c_f}$ is its centroid in mesh coordinates.

For lower bound b_min, extent d, level l, grid width g = 2^l, and small epsilon:

$$\mathbf{u}_f = \mathrm{clip}\left( \left\lfloor g \frac{\mathbf{c}_f - \mathbf{b}_{\min}}{\max(\mathbf{d}, \varepsilon)} \right\rfloor, 0, g - 1 \right)$$

$\mathbf{u_f}$ is the integer cell coordinate; $\mathbf{g = 2^l}$ is grid width at octree level **l**; $\mathbf{b_{min}}$ is the mesh lower bound; **d** is mesh extent; and **epsilon** prevents division by zero on a flat axis.

u_f is the integer cell coordinate. Interleaved coordinate bits form a Morton code. Stable level and code sorting produce reproducible tile numbers. Centroid ownership preserves sum_i n_i = N, where N is the source count. Overlap ownership violates that identity and turns deduplication into a global topology problem.

## 4. Deterministic target allocation

For tile i, source count n_i, floor m_i = min(4,n_i), requested global target T, distributable capacity C, and remaining budget R:

$$C = \sum_i (n_i - m_i), \qquad R = T - \sum_i m_i$$

**$n_i$** is tile **i**'s source count; **$m_i$** is its minimum allocation; **C** is capacity above all floors; **T** is the requested global target; and **R** is the budget remaining after tile floors.

$$q_i = R \frac{n_i - m_i}{C}$$

**$q_i$** is tile **i**'s real-valued proportional quota; **R** is the remaining triangle budget; and **C** is total distributable capacity.

$$t_i = m_i + \lfloor q_i \rfloor + r_i, \qquad r_i \in \{0, 1\}, \qquad \sum_i t_i = T$$

**$t_i$** is the deterministic requested tile target; **floor($q_i$)** is its whole-number quota; and **$r_i$** is the largest-remainder correction assigned with stable tile-order tie breaking.

q_i is the proportional real-valued quota. r_i is assigned in descending fractional-quota order, with tile number as the tie breaker. This largest-remainder construction reaches the requested sum before geometric constraints are applied.

If accepted tile results contain F'_i faces, the actual count and boundary overhead are:

$$T' = \sum_i |F'_i|, \qquad R_B = T' - T$$

**$F'_i$** is the accepted result-face set for tile **i**; **$T'$** is the actual assembled count; **T** is the request; and **$R_B$** is boundary-retention overhead in triangles.

R_B records geometry retained by the boundary contract. On the reference mesh, R_B = 495. A post-pass that deletes these faces would reach an exact counter value by bypassing the constraint that retained them.

## 5. Boundary invariant

Let c_i(e) be the number of tile-i triangles incident on edge e. The protected vertex set is:

$$B_i = \{v : v \in e \text{ and } c_i(e) = 1\}$$

**$c_i(e)$** is edge **e**'s incident-face count inside tile **i**; **v** is an endpoint of an edge with one local incident face; and **$B_i$** is the protected tile-open vertex set.

Every protected source coordinate must appear in the optimized tile result at binary STL precision:

$$\forall v \in B_i, \ \exists v' \in V'_i : \ \text{float32}(\mathbf{x}_{v'}) = \text{float32}(\mathbf{x}_v)$$

**$V'_i$** is the optimized tile vertex set; **$\mathbf{x}_v$** and **$\mathbf{x}_{v'}$** are source and result coordinates; and **float32** applies binary STL precision. Every protected coordinate must survive bit-exactly.

B_i contains vertices on tile-open edges. V'_i is the optimized tile vertex set. Bit-exact float32 comparison matches STL storage and is stricter than a tolerance test.

## 6. Failure analysis

| Attempt | Observed conflict | Resulting decision |
|---|---|---|
| Independent local QEM | Neighboring reducers can move the same seam differently | Freeze tile-open coordinates |
| Overlapping ownership | Duplicate geometry and ambiguous counts | One centroid owner per source face |
| Weld after assembly | Cannot reconstruct differently simplified boundary curves | Preserve before reduction |
| Force exact count | Deletes geometry retained for continuity | Report boundary overhead |
| Completion-order output | Binary result depends on scheduling | Assemble by stable tile number |
| CPU-count workers | Concurrent native state can exceed memory budget | Limit by memory and pending work |
| Trust existing files | Truncated or stale checkpoints can be reused | Validate structure, identity, and digest |
| Write final path directly | Cancellation can expose partial output | Validate temporary file, then replace atomically |

For independently moved seam positions x'_L and x'_R, a crack exists when:

$$\|\mathbf{x}'_L - \mathbf{x}'_R\|_2 > 0$$

$\mathbf{x}'_L$ and $\mathbf{x}'_R$ are independently reduced positions for the same seam point; $\|\cdot\|_2$ is Euclidean distance. Any nonzero value is an open positional disagreement.

A welding tolerance can close some small gaps, but a large tolerance can merge intentional nearby surfaces. It also cannot restore a trajectory already simplified differently on each side.

## 7. Resource, checkpoint, and cancellation model

The persistent index stores raw 50-byte STL triangle records and limits simultaneously open tile streams. Each worker loads one tile, welds equal local coordinates, simplifies it, and atomically writes points, faces, metadata, and a checksum.

$$M_W = \max(64\,\mathrm{MiB},\ 320 n_{\max})$$

**$M_W$** is conservative memory per active worker; **$n_{max}$** is the largest pending tile count; 320 bytes per triangle estimates welded topology, quadrics, adjacency, candidates, and native temporary state.

$$W = \max\left(1, \min\left(W_{\mathrm{configured}}, \left\lfloor \frac{M_{\mathrm{budget}}}{M_W} \right\rfloor, N_P\right)\right)$$

**W** is active worker count; **$W_{configured}$** is the process ceiling; **$M_{budget}$** is available geometry memory; **$M_W$** is estimated memory per worker; and **$N_P$** is pending tile count.

M_W is the conservative per-worker estimate. n_max is the largest pending tile count. M_budget is the geometry-memory budget. N_P is pending work. W is active concurrency. The 320-byte factor covers welded topology, adjacency, quadrics, candidates, and native temporary state.

For checkpoint point bytes P_i and face bytes F_i:

$$H_i = \mathrm{SHA256}(P_i \| F_i)$$

**$P_i$** and **$F_i$** are the serialized point and face arrays for checkpoint **i**; **||** means byte concatenation; and **$H_i$** is the stored SHA-256 content digest.

Reuse also requires matching operation identity, valid array shapes, in-range face indices, and plausible face counts. Cancellation retains complete checkpoints but removes partial assembly output. A resumed run revalidates each tile instead of assuming that file existence means success.

GPU-guided reducers currently use one worker because independent processes can contend for the same selected device and repeat initialization. This is an explicit limitation pending matched tiled-device measurements.

## 8. Deterministic streamed assembly

For stable ascending tile order pi, assembly is:

$$F_{\mathrm{STL}} = F'_{\pi(1)} \,\|\, F'_{\pi(2)} \,\|\, \cdots \,\|\, F'_{\pi(k)}$$

**pi** is stable ascending tile order; $\mathbf{F'_{pi(i)}}$ is one validated tile-face sequence; **k** is tile count; and **||** means ordered record concatenation.

Worker completion order does not appear in the equation. Normals are recomputed for output vertices a, b, and c:

$$\mathbf{n} = \frac{(\mathbf{b}-\mathbf{a}) \times (\mathbf{c}-\mathbf{a})}{\|(\mathbf{b}-\mathbf{a}) \times (\mathbf{c}-\mathbf{a})\|_2}$$

**a**, **b**, and **c** are output triangle vertices; **x** is the cross product; $\mathbf{\|.\|_2}$ is Euclidean norm; and **n** is the recomputed unit normal. Degenerate triangles receive a zero normal.

A degenerate cross product receives a zero normal. Assembly loss and intentional reduction are different quantities:

$$L_{\mathrm{assembly}} = \sum_i |F'_i| - |F_{\mathrm{STL}}|$$

$\mathbf{L_{assembly}}$ is triangles lost during assembly; $\mathbf{|F'_i|}$ is an accepted tile count; and $\mathbf{|F_{STL}|}$ is the final written count. A correct assembly has zero loss.

$$R_S = 1 - \frac{|F_{\mathrm{STL}}|}{|F_{\mathrm{source}}|}$$

$\mathbf{R_S}$ is the intended simplification fraction; $\mathbf{|F_{STL}|}$ is result count; and $\mathbf{|F_{source}|}$ is source count. This is separate from assembly loss.

L_assembly measures whether validated output disappeared during writing. R_S measures intended simplification. All measured runs had L_assembly = 0.

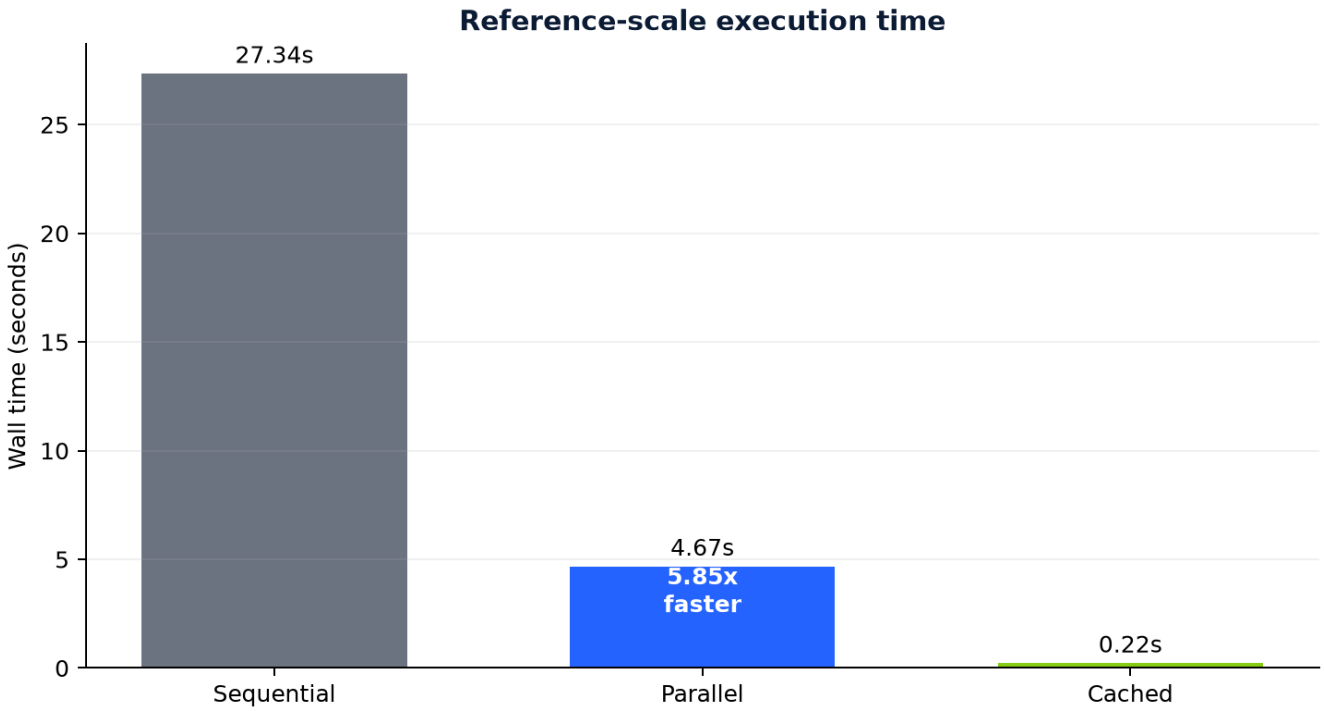
Source fingerprint H_s, target T, algorithm A, and settings G determine the output hash under a deterministic local reducer:

$$(H_s, T, A, G) \longmapsto H_{\mathrm{out}}$$

$\mathbf{H_s}$ is source identity; **T** is requested target; **A** is algorithm identity; **G** is the complete settings set; and $\mathbf{H_{out}}$ is the final output hash.

Stable spatial order, stable quota tie breaking, validated tile identity, and ordered concatenation make scheduling affect runtime without affecting H_out.

# 9. Performance




| Run | Workers | Time | Cached tiles | Output triangles | Assembly loss |
|---|---|---|---|---|---|
| Sequential | 1 | 27.336854 s | 0 | 249,679 | 0 |
| Parallel | 8 | 4.671751 s | 0 | 249,679 | 0 |
| Cached validation and assembly | 1 active | 0.215439 s | 50 | 249,679 | 0 |

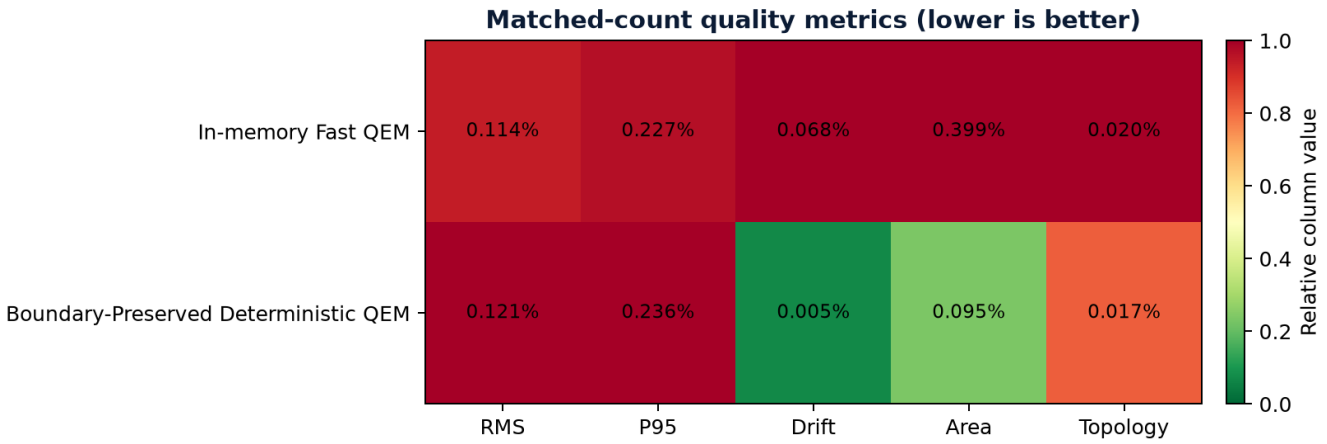
$$S_W = \frac{t_1}{t_W}, \qquad \eta_W = \frac{S_W}{W}$$

**$t_1$** is one-worker time; **$t_W$** is time with **W** workers; **$S_W$** is speedup; and **$\text{eta}_W$** is parallel efficiency on a 0-to-1 scale.

S_W is speedup and eta_W is worker efficiency. The measured S_8 was 5.85 and eta_8 was 0.73. Process startup, storage traffic, unequal tile topology, serialization, and final assembly account for the difference from ideal scaling. The cached trial measures validation and reconstruction, not another simplification pass.

All outputs were 12,484,034 bytes with SHA-256 00455d33955b99149db58e70293484cacbc71372879ea61bcd693abd2133c713.

# 10. Quality measurements




| Measurement | In-memory Fast QEM | Boundary-Preserved Deterministic QEM |
|---|---|---|
| Runtime | 8.581000 s | 4.671751 s |
| Output triangles | 249,184 | 249,679 |
| RMS sampled surface error | 0.113647% | 0.120825% |
| P95 sampled surface error | 0.227477% | 0.235518% |
| Dimension drift | 0.067879% | 0.004893% |
| Surface-area change | 0.399084% | 0.094952% |
| Non-manifold rate | 0.020467% | 0.016855% |

## 11. Quality formulas

For sampled source point x_s, nearest output point y_s, source diagonal D, and sample count S:

$$D_{\mathrm{RMS}} = \frac{100}{D}\sqrt{\frac{1}{S}\sum_{s=1}^{S}\|\mathbf{x}_s - \mathbf{y}_s\|_2^2}$$

$\mathbf{x}_s$ is sampled source point **s**; $\mathbf{y}_s$ is its nearest result point; **S** is sample count; **D** is source diagonal; and $\mathbf{D}_{\mathbf{RMS}}$ is average normalized surface displacement in percent.

$$D_{95} = \frac{100}{D}\,\mathrm{percentile}_{95}(\|\mathbf{x}_s - \mathbf{y}_s\|_2)$$

$\mathbf{D}_{\mathbf{95}}$ is the normalized 95th-percentile sampled displacement; $\mathbf{x}_{\mathbf{s}}$, $\mathbf{y}_{\mathbf{s}}$, and **D** have the same meanings as above. It exposes the higher-error tail that an average can hide.

D_RMS measures average sampled displacement. D_95 exposes the higher-error tail. Normalizing by D makes the result comparable across model scales.

For source extent d_j, output extent d'_j, source area A, output area A', output edge incidence c'(e), and output faces F':

$$\delta_j = 100\frac{|d'_j - d_j|}{D}$$

$\mathbf{d}_{\mathbf{j}}$ and $\mathbf{d}'_{\mathbf{j}}$ are source and result extents on axis **j**; **D** is source diagonal; and $\mathbf{delta}_{\mathbf{j}}$ is normalized dimension drift in percent.

$$D_A = 100\frac{|A' - A|}{A}$$

**A** and $\mathbf{A}'$ are source and result surface areas; $\mathbf{D}_{\mathbf{A}}$ is absolute relative area change in percent.

$$D_N = 100\frac{|\{e : c'(e) > 2\}|}{|F'|}$$

$\mathbf{c}'\mathbf{(e)}$ is result edge incidence; edges above two are non-manifold; $\mathbf{|F'|}$ is result triangle count; and $\mathbf{D}_{\mathbf{N}}$ is the normalized non-manifold rate in percent.

delta_j measures bounding-dimension drift on axis j. D_A measures surface-area change. D_N measures non-manifold edge incidence relative to result size. No single value proves quality. RMS can hide isolated damage, P95 can miss rare extremes, bounds can remain stable while interior detail changes, and topology can improve while distance rises slightly. The metrics are interpreted together.

On the reference mesh, the tiled result traded 0.007178 percentage points of RMS and 0.008041 points of P95 error for lower drift, area change, non-manifold rate, and runtime. These values characterize this fixture and machine, not every mesh class.

# 12. Validation contracts

**Validated production contracts**

| Contract | Result | Evidence |
|---|---|---|
| Source Partition | PASS | sum of tile source counts equals 4126315 |
| Deterministic Target | PASS | repeated largest-remainder allocation is identical and sums to requested target |
| Assembly Count | PASS | assembled record count equals 249679 validated tile triangles |
| Assembly Payload | PASS | assembled triangle payload equals ordered tile payload concatenation |
| Parallel Consistency | PASS | sequential and parallel SHA-256 values match |
| Checkpoint Consistency | PASS | cached reconstruction SHA-256 matches fresh runs |
| Connected Seams | PASS | source and output boundary-edge counts match |
| Cancellation | PASS | no published STL or partial file remains |
| Resume | PASS | completed checkpoint reused and final hash matches |
| Corruption | PASS | invalid checkpoint rejected and rebuilt |

A 260 by 130 quad grid produces 67,600 triangles and crosses several tiles. With exact-coordinate welding, boundary-edge count is:

$$N_B = |\{e : c(e) = 1\}|$$

**c(e)** is exact-coordinate welded edge incidence; **e** is an undirected edge; and $\mathbf{N_B}$ is the number of boundary edges. Matching source and output values detects newly opened seams on the connected fixture.

The contract requires N_B_output = N_B_source. This detects newly opened seams on the connected fixture. It does not claim to repair openings already present in the source.

| Contract | Observed result |
|---|---|
| Source ownership | Every source triangle assigned once |
| Assembly count | Equals sum of validated tile outputs |
| Artificial open seams | 0 detected |
| Sequential versus parallel | Byte-identical |
| Fresh versus cached | Byte-identical |
| Cancellation | No final or partial STL published |
| Resume | Completed checkpoints reused |
| Corrupt checkpoint | Rejected and rebuilt |

## 13. Evidence limits and future work

The evidence establishes deterministic bytes across tested schedules, zero assembly loss, cancellation cleanup, resumable validated checkpoints, and no new open edges on the connected fixture. It does not prove watertightness, manifoldness, absence of self-intersections, or universal quality for arbitrary inputs.

Boundary retention is conservative. A future shared-boundary optimizer could reduce R_B, but it needs the same deterministic identity, cancellation, and recovery contracts. Additional adversarial fixtures should include curved seams, thin shells, intersecting layers, pre-existing non-manifold edges, disconnected components, degenerate triangles, and extreme coordinate ranges.

The current index targets binary STL. Formats with shared normals, colors, materials, or texture coordinates require streamed readers and explicit attribute-preservation rules. A complete source hash can supplement the bounded fast invalidation fingerprint when cryptographic source identity is required.

## 14. Prior work and attribution

Garland and Heckbert established iterative pair contraction with accumulated quadric error matrices. Lindstrom established out-of-core simplification using clustering and quadric placement. Cignoni and colleagues described external-memory octree processing with protected block-boundary vertices. Their work supplied essential mathematical and systems foundations. This report focuses on the production integration and the measured contracts around those foundations.

1. M. Garland and P. Heckbert. Surface Simplification Using Quadric Error Metrics. SIGGRAPH 1997.
2. P. Lindstrom. Out-of-Core Simplification of Large Polygonal Models. SIGGRAPH 2000.
3. P. Cignoni et al. External Memory Management and Simplification of Huge Meshes.

## 15. Reproduction

The publication package includes a frozen implementation snapshot, contract tests, benchmark driver, normalized CSV and JSON evidence, figure generator, experiment catalog, attribution record, and detailed algorithm and design-decision notes. The private reference mesh is not distributed.

```
python benchmark_boundary_preserved_deterministic_qem.py INPUT.stl --target 249184 --workers 8
--output OUTPUT_DIRECTORY
```

The benchmark clears result state between fresh sequential and parallel runs, retains checkpoints for the cached run, emits machine-readable results, and fails if output hashes or assembled counts disagree.

# Independent Research, Related Work, and Attribution

Make Your Own World R&D

The purpose of this project is to explore ideas, solve problems, document what we learn, and make useful work available for others to build upon.

Our work may involve original developments, improvements to established methods, independent rediscovery, or approaches that overlap with existing research and projects. We consider all of these worthwhile contributions. Novelty, priority, and recognition are not prerequisites for useful work.

We recognize that virtually all research and development builds upon generations of work by others. We make a good-faith effort to identify and acknowledge relevant publications, projects, tools, and contributors, and to respect applicable licenses and attribution requirements.

Some related work may only become known after research or development has already taken place. Being cited as related work does not necessarily mean that it influenced or was used in the original implementation. Where external work is directly incorporated or adapted, we aim to identify and credit it appropriately.

No individual or project can reasonably guarantee awareness of every related idea, publication, implementation, or independent discovery. References and acknowledgments therefore reflect the information available at the time and may be expanded or corrected as additional work comes to our attention.

We do not claim exclusive ownership of general ideas, methods, or discoveries, nor do we assume that independent development establishes historical priority. Similarity or overlap should not, by itself, be interpreted as evidence of copying or misappropriation.

We welcome constructive corrections, additional references, and contributions that improve the accuracy and usefulness of the published record. Our focus is on the work itself and the knowledge it makes available, rather than disputes over recognition or precedence.

**Whether something is new, familiar, independently rediscovered, or simply useful, our goal is the same: preserve what was learned, share it openly, and help others move forward.**